

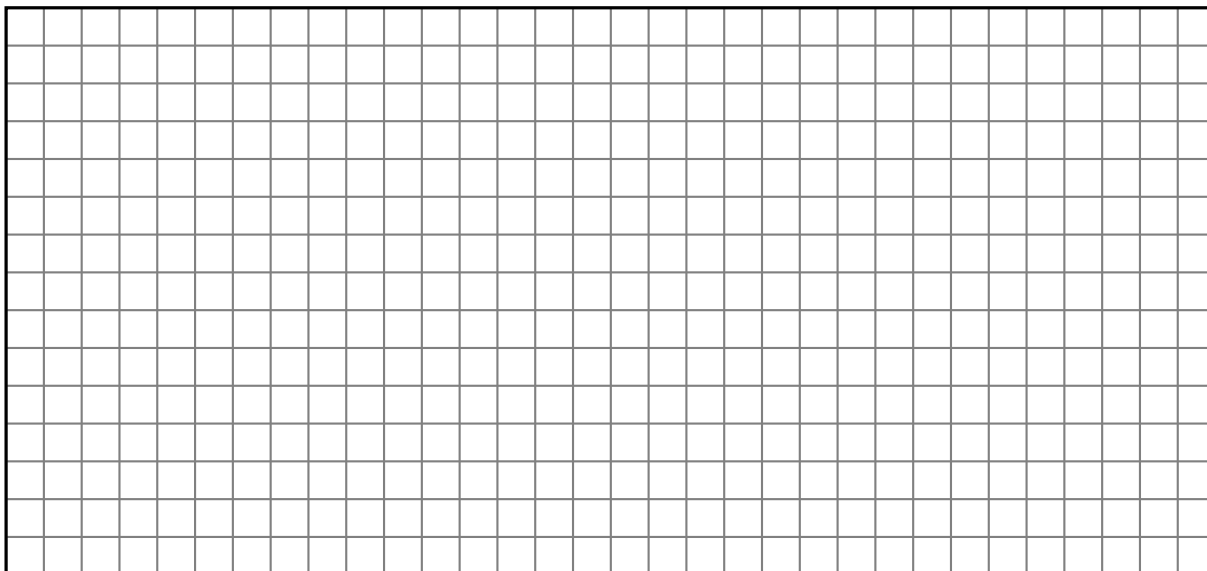
Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (24 points)*Submit the postlab to the TA at the end of the lab.***Experiment 1 (12 points)****Question 4****3 points**

For each h_i recorded during the experiment, calculate the corresponding t_i in milliseconds and enter it in the table below. Pay attention to units.

Trial	Your Data		Partner's Data		Trial	Your Data		Partner's Data	
	h [cm]	t [ms]	h	t		h [cm]	t [ms]	h	t
1					14				
2					15				
3					16				
4					17				
5					18				
6					19				
7					20				
8					21				
9					22				
10					23				
11					24				
12					25				
13									

**Question 5****3 points**

Use the graph above to draw cumulative distribution functions for both your and your partner's reaction times. Choose an appropriate bin size based on your data. Include axes, labels, units, and a title. In future plots you will not be reminded to do this.

Question 6**3 points**

Calculate the mean and standard deviation for each of the two data sets you collected. Do not forget units. Write your results in the form $\hat{\mu} \pm \hat{\sigma}$:

	Your Values	Partner's Values
Height	$\pm$	$\pm$
Time	$\pm$	$\pm$

Question 7**3 points**

Use error propagation to compute the variance in your t measurements. How does it compare to the standard deviation you computed in the previous question?

Experiment 2 (12 points)**Question 8****2 points**

Record your Geiger counter readings in the table below and sketch the inside of the steel box.

Reading	No Plate	0.12 mm Plate	0.60 mm Plate
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
Total			

Sketch of the inside of the steel box:

Question 9**2 points**

Consider only your first reading from each of the three setups. What is the statistical error in each case? Does this single reading distinguish between the three setups? Explain.

Question 10**2 points**

Sum all ten readings for each setup and enter the totals in the last row of your table. What is the statistical error in each total? Does the combined data distinguish between the three setups? Explain.

Question 11**2 points**

For each setup, count how many of your ten readings fall within one error bar of the mean, and how many fall within two error bars. Organize your results in a table.

Question 12**3 points**

Compare the error from a single reading to the error from the total of ten readings. Is their ratio what you expect? Explain. What errors would you expect for 100 readings? For 1000?

Question 13**1 point**

How many readings would you need to unambiguously distinguish between the no-plate and 0.12 mm-plate setups? Explain your reasoning.